Algebra 1 Name Answer Key
Unit 5 Date____________
Lesson 9 Homework Period____________

1. Solve the system of equations using elimination.

$$+\begin{cases}3x+6y=18\\-3x+4y=2\end{cases}$$

$10y = 20$
$y = 2$

$3x + 6(2) = 18$
$3x + 12 = 18$
$3x = 6$
$x = 2$

$(2,2)$

2. Verify the solution from question 1 algebraically using substitution.

$3(2) + 6(2) = 18$
$6 + 12 = 18$
$18 = 18$ ✓

$-3(2) + 4(2) = 2$
$-6 + 8 = 2$
$2 = 2$ ✓

3. Solve the system of equations using elimination.

$$+\begin{cases}2x-7y=10\\4x+7y=20\end{cases}$$

$6x = 30$
$x = 5$

$2(5) - 7y = 10$
$10 - 7y = 10$
$-7y = 0$
$y = 0$

$(5,0)$

4. Verify the solution from question 3 algebraically using substitution.

$2(5) - 7(0) = 10$
$10 = 10$ ✓

$4(5) + 7(0) = 20$
$20 = 20$ ✓

5. Solve the system of equations using elimination.

$$\begin{cases}2x-4y=-12\\x+3y=9\end{cases}$$

$2x - 4y = -12$
$-2x - 6y = -18$

$-10y = -30$
$y = 3$

$2x - 4(3) = -12$
$2x - 12 = -12$
$2x = 0$
$x = 0$

$(0,3)$

6. Verify the solution from question 5 algebraically using substitution.

$2(0) - 4(3) = -12$
$-12 = -12$ ✓

$(0) + 3(3) = 9$
$9 = 9$ ✓

7. Solve the system of equations using elimination.

$$\begin{cases} (3x - 4y = -14)2 \\ (2x - 6y = 8) - 3 \end{cases}$$

$$\begin{aligned} 6x - 8y &= -28 \\ \underline{-6x + 18y} &\underline{= -24} \\ 10y &= -52 \\ y &= -5.2 \end{aligned}$$

$$\begin{aligned} 2x - 6(-5.2) &= 8 \\ 2x + 31.2 &= 8 \\ 2x &= -23.2 \\ x &= -11.6 \end{aligned}$$

$(-11.6, -5.2)$

8. Verify the solution from question 7 algebraically using substitution.

$$\begin{aligned} 3(-11.6) - 4(-5.2) &= -14 \\ -34.8 + 20.8 &= -14 \\ -14 &= -14 \checkmark \end{aligned}$$

$$\begin{aligned} 2(-11.6) - 6(-5.2) &= 8 \\ -23.2 + 31.2 &= 8 \\ 8 &= 8 \checkmark \end{aligned}$$

9. Solve the system of equations using elimination.

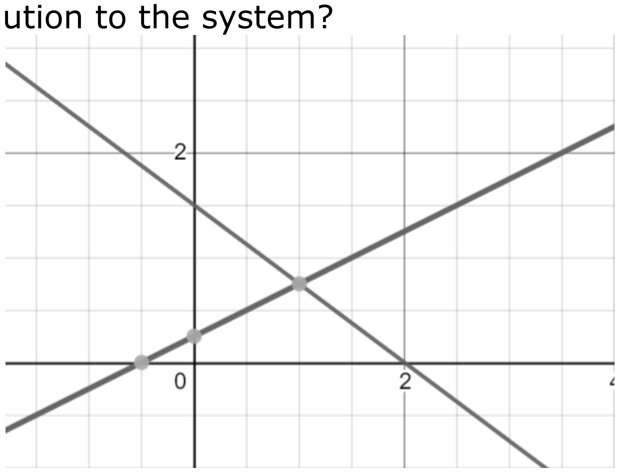
$$- \begin{cases} 3x + 4y = 6 \\ -2x + 4y = 1 \end{cases}$$

$$\begin{aligned} 5x &= 5 \\ x &= 1 \end{aligned}$$

$$\begin{aligned} -2(1) + 4y &= 1 \\ -2 + 4y &= 1 \\ 4y &= 3 \\ y &= \tfrac{3}{4} \end{aligned}$$

$\left(1, \frac{3}{4}\right)$

10. The graph of the system from question 9 is shown below. How can the graph be used to verify the solution to the system?



Look at the point of intersection. Here, $\left(1, \frac{3}{4}\right)$ is the point of intersection, verifying the solution is correct.